Salary > 8 lakhs ?

Yes

Credit score > 700

Yes

Decision Tree

Yes	No	Entropy	Gini
10	0	0	0
8	2	0.7	0.32
5	5	1	0.5

0 - 0.5

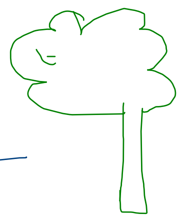
ID3 , C4.5 , CART

Dataset
↓
Find best split
↓
left subset → Find best split → Repeat
right subset → Find best split → Repeat

Recursive Binary Splitting.

How does the computer know which question is best?

- Root node
- internal node
- branch.
- Leaf node
- Depth.



Pure

Stud.	Pass	
A	Yes	Y
B	Yes	N
C	Yes	Y
D	Yes	N
E	Yes	N

→ uncertainty is zero.
→ perfectly pure.

More mixed = more uncertainty
Less mixed = more purity.

Entropy

$$y = -x\log_2(x) - (1-x)\log_2(1-x)$$

high entropy → data is highly mixed
low " → data is almost pure.

Case 1. Yes 10, No 0 — perfectly pure node. entropy = 0

Case 2. Balanced node — Yes 5, No 5 — ent = 1, highly uncertain.

Case 3. Yes 8, No 2 — Pure ---------- Mixed: 0 … 0.4 0.5 0.7 … 1

10 students → Pass / Fail
- Hours studied
- Attendance.
- assignments submi

Information Gain

IG

	H.S.	
A	8	Pass
B	7	Pass
C	6	Pass
D	7	Pass
E	8	;
F	7	;

	H.S.	
G	2	Fail
H	3	Fail
I	4	Fail
J	1	Fail
K	2	Fail

Color
shape.

IG = Entropy before split — Entropy after split.

Gini :- squared probabilities.

Gini = 0 → record belong to same class

$= 1 - \sum p_i^2$

0.64 + 0.04 = 0.68

0 ——— 1 (0.32), pure.

$= 1 - (0.8)^2 - (0.2)^2 = 1 - 0.68 = 0.32$